

Rock Hardness: Mark or Groove?

I can explain relative hardness and identify a fair, safe scratch method for comparing rock samples.

MARK

do not decide yet

WIPE

check safely

GROOVE

relative hardness

[Open the original interactive lesson](#)

What success looks like

- I can say that hardness means resistance to scratching.
- I can distinguish a loose tool streak from a groove that remains after wiping.
- I can change the sample while keeping the tool, passes and guide as consistent as practical.

Retrieve and reconnect

- Explain why the W9 sorting question stayed unchanged.
- Name one visible rock property and one property that needs a test.
- State why matching units makes a comparison fair.
- Lead-in: what must be checked before a pale line is called a scratch?

A pale line appears after a secured blunt tool passes over a rock. What should happen before we call it a scratch groove?

Think first. Point, say, sign or write your choice. Give one reason.

- A. Wipe the line and inspect whether an indentation remains.
- B. Weigh the rock and call the heaviest one hardest.
- C. Choose harder until every rock shows a deep line.

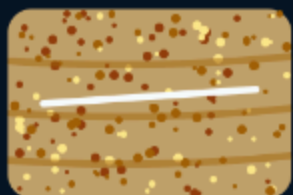
Look closely · what is the evidence?

PASS 1 · a pale line appears on the dry surface

WIPE · loose tool material is removed

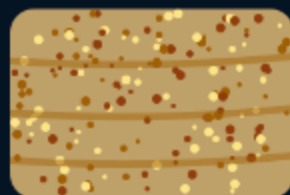
OBSERVE · an indentation remains, or the line disappears

Look closely · what is the evidence?



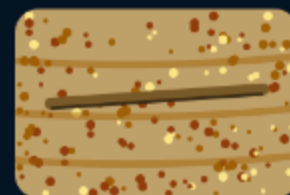
PASS 1 · a pale line appears

A rock with a pale line after one tool pass



WIPE · loose tool dust removed

The same rock wiped clean



groove REMAINS · softer line GONE · harder

Two rocks after wiping: one keeps a groove, one does not

Words we will use

- hardness — resistance to a scratch
- groove — a line cut into the surface
- control — what we keep the same

Watch the model

- Place a teacher-pretested dry sample in a stable tray. Use only the secured blunt scratch jig or a teacher demonstration.
- Use the same tool, guide length, number of passes and safe force proxy. Never choose harder to make an expected result appear.
- Wipe the line with the prepared method, then inspect from the same viewing position for a remaining groove.

Watch the model



Scratch test on the chosen rock

The pale line wipes away and no indentation remains. Which record is accurate?

- A. No groove remained; this sample resisted the stated scratch method.
- B. A deep groove formed, so the rock is soft.
- C. The rock is unbreakable in every situation.

Commit to an answer. Show the evidence you used. Listen, then revise if needed.

Build the explanation

- Compare outcomes with cautious words: deep groove, shallow groove, faint groove or no groove under this method.
- A clearer remaining groove means lower resistance to this particular scratch method; it does not measure every kind of strength.
- If a sample flakes, crumbles or makes dust, stop. Preserve the observation and use the representative digital or image route rather than forcing a repeat.

Find and repair the misconception

Hardness glitch: the heaviest rock is the hardest and cannot break.

A. Keep it: weight, hardness and strength are the same.

B. Repair it: hardness is resistance to scratching under the stated method.

C. Repair it: break the rocks to see which is hardest.

Mark, Wipe or Groove Evidence Gate

Place each method or observation under the role it performs in a valid hardness comparison.

- Lock the stated method.
- Commit a prediction.
- Make or reveal the safe tool pass.
- Wipe and inspect.
- Record the bounded result.

Mark, Wipe or Groove Evidence Gate

Use the numbered cards in your pupil pack. Take turns to choose, justify and check. You may direct a partner to move a card.

Choose a group · defend your placement

- VALID GROOVE EVIDENCE
- VALID NO-GROOVE EVIDENCE
- KEEP THIS THE SAME
- DOES NOT MEASURE HARDNESS
- STOP / USE PARITY ROUTE

Name what changed, three controls and the observation that decides groove or no groove.

Your independent mission

Plan and direct a safe two-sample relative hardness comparison, then record the method, groove observation and bounded conclusion.

Record: A Rock Hardness Method Card with prediction, changed sample, controls, after-wipe observation and a relative conclusion within the stated method.

Choose your support and challenge

- Supported: Choose groove/no-groove after-wipe cards for two samples and direct the adult through the same method.
- Standard: Compare two coded samples with the same stated method and justify which resisted a groove more.
- Stretch: Evaluate the controls, distinguish hardness from strength and propose one safe repeat or method improvement.

Show what you know

Explain how the method distinguishes a tool streak from a groove and name two controls.

Point to, read or explain the evidence you made. Choose one next step after the adult has listened.